0. Introduction to categories

September 21, 2014

These are notes for Algebra 504-5-6. Basic category theory is very simple, and in principle one could read these notes right away—but only at the risk of being buried in an avalanche of abstraction. To get started, it's enough to understand the definition of "category" and "functor". Then return to the notes as necessary over the course of the year. The more examples you know, the easier it gets.

Since this is an algebra course, the vast majority of the examples will be algebraic in nature. Occasionally I'll throw in some topological examples, but for the purposes of the course these can be omitted.

1 Definition and examples of a category

A *category* $\mathcal{C}$ consists first of all of a class of objects $Ob\mathcal{C}$ and for each pair of objects X, Y a set of *morphisms* $Mor_{\mathcal{C}}(X, Y)$. The elements of $Mor_{\mathcal{C}}(X, Y)$ are denoted $f : X \longrightarrow Y$, where f is the morphism, X is the *source* and Y is the *target*. For each triple of objects X, Y, Z there is a *composition law*

$$Mor_{\mathcal{C}}(X, Y) \times Mor_{\mathcal{C}}(Y, Z) \longrightarrow Mor_{\mathcal{C}}(X, Z),$$

denoted by $(\phi, \psi) \mapsto \psi \circ \phi$ and subject to the following two conditions:

(i) (associativity) $h \circ (g \circ f) = (h \circ g) \circ f$

(ii) (identity) for every object X there is given an *identity morphism* Id_X satisfying $Id_X \circ f = f$ for all $f : Y \longrightarrow X$ and $g \circ Id_X = g$ for all $g : X \longrightarrow Y$.

Note that the identity morphism Id_X is the unique morphism satisfying (ii). Very frequently we will use the alternative notation $Hom_{\mathcal{C}}(X, Y)$ ("Hom" for "homomorphism") in place of Mor because it fits well with our algebraic examples. Or we may use more traditional notations such as $Hom_F(V, W)$ for the F-linear maps $V \longrightarrow W$.

A few examples make these concepts transparent. In all of the following the composition law is ordinary composition of functions, and the identity is the usual identity function. The category is given a name in boldface, followed by a description of its objects and morphisms.

- **Set**: sets and functions between them

- **Grp**: groups and group homomorphisms
- **Rng**: rings and ring homomorphisms
- **F-vect**: F-vector spaces and F-linear maps (F a given field)
- **Top**: topological spaces and continuous maps

In each case one has to check that the functions under consideration (group homomorphisms, continuous maps, etc.) are closed under composition, in order to define the composition law in the usual way. But this is completely trivial. The verification of conditions (i)-(ii) in these examples is equally trivial, a common state of affairs in category theory. Occasionally one encounters examples where there is something to prove; for example if $\mathcal{C}$ is the category of open subsets of some $\mathbb{R}^n$ ($n \geq 1$ is allowed to vary) with morphisms the differentiable maps, then to define the composition law one needs to know that the composition of differentiable maps is differentiable, i.e. one must prove the chain rule.

In the examples above, and in most of the examples we will consider, the objects of the category are "sets with some extra structure", and the morphisms are "maps of sets that preserve the structure" in some specified sense. Let us informally call categories of this type "concrete". But there is nothing in the definition that requires such concreteness, and indeed the concept of a category is vastly more general. Here are two examples that we may occasionally use, plus a topological example for those who are so inclined:

Example. Let M be a monoid. Then M defines a category with one object X and $Mor\,(X,X) = M$; the composition law is given by the multiplication on M. Conversely a category with one object defines a monoid, by the same construction. Thus a category can be regarded as a (vast!) generalization of a monoid. The case when M is a group is of particular interest; we will return to this shortly.

Example. Let S be a partially ordered set. Then S defines a category whose objects are the points of S, and such that $Mor\,(x,y)$ has cardinality 1 if $x \leq y$ and is empty otherwise. The composition law comes from transitivity of the order, and identities from $x \leq x$. In fact this reinterpretation of S amounts to little more than a change of notation, writing $x \longrightarrow y$ in place of $x \leq y$.

Example: the homotopy category. If X, Y are topological spaces, two continuous maps $f_0, f_1 : X \longrightarrow Y$ are said to be *homotopic* if there is a continuous map $H : X \times [0,1] \longrightarrow Y$ such that H restricted to $X \times \{i\}$ is f_i for $i = 0, 1$. This defines an equivalence relation on the set of continuous maps $X \longrightarrow Y$; the equivalence classes are called *homotopy classes*. The *homotopy category* **hTop** is the category whose objects are topological spaces and whose morphisms are homotopy classes of maps (here one has to check that composition is well-defined on homotopy classes, but this is easy).

A morphism $f : X \longrightarrow Y$ in a category is an *isomorphism* if there is a morphism $g : Y \longrightarrow X$ such that $g \circ f = Id_X$ and $f \circ g = Id_Y$. It is easily checked that the *inverse* g is unique; sometimes we denote it f^{-1}. In the examples **Set**, **Grp**, **Rng**, **Top**, the isomorphisms are

respectively the bijections, group isomorphisms, ring isomorphisms, and homeomorphisms. In **hTop** the isomorphisms are called *homotopy equivalences*.

Caution: It needn't be true that an isomorphism is the same thing as a bijective morphism. In fact such an assertion doesn't even make sense in general, since the objects of the category need not be sets, and even if they are, the morphisms need not be functions (the homotopy category is a good example; injectivity and surjectivity are not invariant under homotopy, and therefore have no meaning in **hTop**). Furthermore, even in a "concrete" category bijective morphisms need not be isomorphism. For example, in **Top** it is well-known that a continuous bijection need not be a homeomorphism. In the algebraic categories we'll be working in, however, it is often the case that an isomorphism is the same thing as a bijective morphism.

An *endomorphism* of an object X is a morphism $X \longrightarrow X$. An *automorphism* is an isomorphism $X \longrightarrow X$.

A category in which every morphism is an isomorphism is called a *groupoid*. Note that in the monoid example discussed above, if $M = G$ is a group, then the associated category is a groupoid with one object. Conversely any groupoid with one object X is the same thing as a group, since it is determined by the group $G := Mor\,(X, X)$.

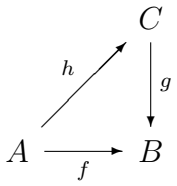
A topological example: the fundamental groupoid of a space. Like all of our topological examples, this one is optional; if it doesn't make sense, ignore it. Let X be a topological space. The objects of the fundamental groupoid $\pi(X)$ are the points of X. A morphism $x \longrightarrow y$ is a path-homotopy class of paths from x to y. Here the term "path-homotopy" indicates that homotopies between paths are required to be stationary on the endpoints. Morphisms are composed by simply concatenating the two paths. (*Caution*: The usual way of writing concatenations is $\lambda * \mu$ for the path that follows first λ, then μ. In our categorical notation we prefer to write $\mu \circ \lambda$ for first λ, then μ.) One interesting feature here is that concatenation is not an associative operation on the nose; it only becomes associative after passing to path-homotopy classes. The identity Id_x is the constant path at x. Every morphism is an isomorphism, because we can "invert" a path by running it in reverse. Hence $\pi(X)$ is a groupoid. Needless to say there are numerous points to be checked to make all this precise. The automorphism group of an object x is the "fundamental group based at x" (subject to the *caution* above).

Remark. In general, the morphism sets (or "hom sets") in a category are only sets. But in some cases, including many of the cases important to us, $Hom_{\mathcal{C}}(X, Y)$ has an additional structure. For example in **Ab**, for any abelian groups A, B we know that $Hom\,(A, B)$ is itself an abelian group. Moreover if $f : B \longrightarrow C$ is a group homomorphism then the post-composition map $Hom\,(A, B) \longrightarrow Hom\,(A, C)$ given by $g \mapsto f \circ g$ is a group homomorphism, and similarly for precomposition. Similarly in **F-Vect**, we know that $Hom_F(V, W)$ is an F-vector space, etc. We'll see more examples throughout the course.

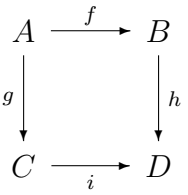
2 Commutative diagrams

Commutative diagrams constitute the greatest notational advance since the decimal system. Before long, you won't be able to imagine how you ever lived without them.

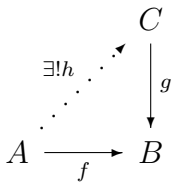
Suppose given objects A, B, C in a category $\mathcal{A}$ (picture your favorite category!), and morphisms $f : A \longrightarrow B$, $g : C \longrightarrow B$, $h : A \longrightarrow C$ such that $g \circ h = f$. We can display this information in a *commutative diagram*, in this case a *commutative triangle*,



where "commutative" or "the diagram commutes" means precisely that $g \circ h = f$. Similarly a *commutative square* is a diagram (the vertices are objects in the category; the arrows are morphisms)

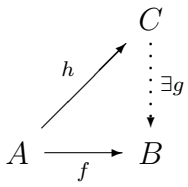


such that $h \circ f = i \circ g$. Thus all the "commutative diagram" says is that we have the four indicated morphisms and that two composites going from A to D are equal. Even though it takes up more space than simply writing the equation, the use of commutative diagrams to display equality of compositions is fantastically useful. The point is that you can see at a glance the source and target of each morphism, and how the compositions are related. A equally useful refinement of this notation incorporates dotted arrows to distinguish morphisms that are given (the solid arrows) from morphisms that are asserted to exist (the dotted arrows). For example, in the original triangle setup above, suppose we are only given the morphisms f and g, but we have proved or wish to prove "there is a unique $h : A \longrightarrow C$ such that $g \circ h = f$". All of this data is neatly displayed in the commutative diagram



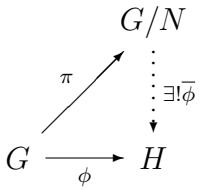
Thus f, g are given, and we are asserting the existence of a unique h, sometimes called a "lift", commuting in the diagram.

Equally common is the situation where f, h are given, and we want to assert the existence (and perhaps uniqueness) of a morphism $g : C \longrightarrow B$ such that $g \circ h = f$. This data is displayed by a commutative diagram



Again the solid arrows are given, and we are asserting the existence of the dotted arrow g (by the way, "arrow" is often used synonymously with "morphism"). If we are claiming a unique g we would write $\exists! g$. A useful term here is to say that "f factors through C" or "f factors uniquely through C".

Example. Suppose $\phi : G \longrightarrow H$ is a group homomorphism, and N is a normal subgroup of G. Let $\pi : G \longrightarrow G/N$ denote the quotient homomorphism. Then by elementary group theory we know that if $N \subset Ker\,\phi$, then ϕ factors uniquely through π. Diagramatically:



Finally, we note that the concept "commutative diagram" extends to diagrams built on any directed graph; triangles and squares are just the cases that most commonly occur.

3 New categories from old

3.1 Subcategories

Let $\mathcal{C}$ be a category. A *subcategory* $\mathcal{D}$ of $\mathcal{C}$ is a category whose objects are objects of $\mathcal{C}$ and such that for all objects X, Y of $\mathcal{D}$, $Mor_{\mathcal{D}}(X, Y) \subset Mor_{\mathcal{C}}(X, Y)$, with the composition law obtained by restricting the composition law in $\mathcal{C}$, and with the same identities. We call $\mathcal{D}$ a *full* subcategory if $Mor_{\mathcal{D}}(X, Y) = Mor_{\mathcal{C}}(X, Y)$; in other words, we have restricted the objects but not the morphisms. Some examples of full subcategories:

- **fSet**, the full subcategory of finite sets in **Set**
- **Ab**, the full subcategory of abelian groups in **Grp**
- **cRing**, the full subcategory of commutative rings in **Rng**
- **chTop**, the full subcategory of compact Hausdorff spaces in **Top**.

At the opposite extreme, one could keep the same objects but restrict the morphisms in such a way that the restricted morphisms are closed under composition. For example, if $\mathcal{C}$ is any category we can form a subcategory $\mathcal{C}_{iso}$ with the same objects but with morphisms the isomorphisms (it is clear that isomorphisms are closed under composition).

3.2 Product categories

Suppose given categories $\mathcal{A}$, $\mathcal{B}$. The *product category* $\mathcal{A} \times \mathcal{B}$ has objects the pairs (A, B) with A, B objects of $\mathcal{A}$, $\mathcal{B}$ respectively, and morphisms $(A_1, B_1) \longrightarrow (A_2, B_2)$ given by a pair of morphisms $f : A_1 \longrightarrow A_2$, $g : B_1 \longrightarrow B_2$. One reason for introducing product categories is to have a natural home for "functors of two variables" such as the cartesian product of two sets, groups, rings etc., as will be explained below.

More generally one could take any index set I and collection of categories $\mathcal{A}_i$ for $i \in I$, and form a product category $\prod_{i \in I} \mathcal{A}_i$ in the analogous way.

3.3 Morphism categories

Let $\mathcal{C}$ be any category. We can form a new category $Mor\,\mathcal{C}$ whose objects are morphisms $f : X \longrightarrow Y$ and whose morphisms are commutative squares; that is, a morphism from $f_1 : X_1 \longrightarrow Y_1$ to $f_2 : X_2 \longrightarrow Y_2$ is a commutative square

$$\begin{array}{ccc} X_1 & \xrightarrow{f_1} & Y_1 \\ \downarrow & & \downarrow \\ X_2 & \xrightarrow[f_2]{} & Y_2 \end{array}$$

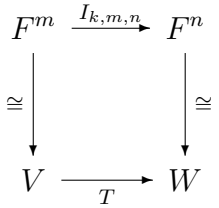
Composition of squares is defined in the evident way (glue the squares together along their common edge), and we have our category. Incidentally, in the diagram above I chose not to label the vertical arrows. They are morphisms in our category, but for the moment they prefer to remain anonymous, rather than add to notational clutter.

In particular, when we say that two morphisms in a category are isomorphic, we mean isomorphic in the category $Mor\,\mathcal{C}$. Abstract though it may appear, this notion is in fact very concrete, as I will illustrate with an example from elementary linear algebra. Let F be a field—any field, choose your favorite—and let V, W be vector spaces of dimension m, n respectively over F. Let $T : V \longrightarrow W$ be a linear transformation of rank k. Then a result of basic linear algebra says that we can choose bases $e_1, ..., e_m$ for V and $f_1, ..., f_n$ for W such that the $n \times m$ matrix A_k of T with respect to these bases has the block form

$$\begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix}$$

Here I_k is the $k \times k$ identity matrix, and the 0-blocks are of the appropriate sizes to fill out the matrix. Let's denote this matrix $I_{k,m,n}$. Often this result is presented in an even more elementary form, with $V = F^m$ and $W = F^n$, so that T starts off in life as a rank k matrix, and the theorem asserts T is "equivalent" to the matrix above.

But another way to look at it is that we have a commutative diagram of linear maps (identifying the matrix $I_{k,n,m}$ with a linear transformation $F^m \longrightarrow F^n$)



where the vertical isomorphisms arise from the bases chosen. In other words, the assertion is that T is isomorphic to $I_{k,m,n}$ in the category $Mor\,(F-Vect)$ of morphisms of F-vector spaces, and consequently we have classified all such morphisms (at least for finite-dimensional vector spaces) up to "isomorphism of morphisms". We will meet many such classification theorems.

4 Functors

4.1 Covariant functors

Let $\mathcal{A}$, $\mathcal{B}$ be categories. A *covariant functor* $F : \mathcal{A} \longrightarrow \mathcal{B}$ assigns to each object A of $\mathcal{A}$ an object $F(A)$ of $\mathcal{B}$, and to each pair of objects A_1, A_2 of $\mathcal{A}$ a function

$$F : Hom_{\mathcal{A}}(A_1, A_2) \longrightarrow Hom_{\mathcal{B}}(F(A_1), F(A_2))$$

such that

(i) F commutes with composition: $F(g \circ f) = F(g) \circ F(f)$

(ii) $F(Id_A) = Id_{F(A)}$.

Notes: 1. If $f : A_1 \longrightarrow A_2$ is a morphism in $\mathcal{A}$, the induced morphism $F(f) : F(A_1) \longrightarrow F(A_2)$ is sometimes denoted f_*, assuming the functor F in question is understood.

2. Functors come in two flavors, covariant and contravariant, with the latter type to be discussed shortly. However, we often refer to covariant functors simply as functors, when no confusion can result.

3. It is immediate from the definitions that functors preserve commutative diagrams.

In practice, the verification of conditions (i)-(ii) is so trivial that we don't even bother to mention it; in fact often we don't even bother to mention the equally obvious definition of the induced morphisms.

Example. We might say "Let $F : \mathbf{Rng} \longrightarrow \mathbf{Grp}$ be the functor sending a ring R to its group of units $R^\times$", with no further explanation. Here any ring homomorphism $\phi : R \longrightarrow S$ takes units to units, so defines a group homomorphism $\phi^\times : R^\times \longrightarrow S^\times$; the implicit definition is $F(\phi) = \phi^\times$, and conditions (i)-(ii) are obvious.

Example. Define a functor $t : \mathbf{Ab} \longrightarrow \mathbf{Ab}$ by sending an abelian group A to its torsion subgroup tA (i.e. the subgroup of elements of finite order; it is a subgroup thanks to the abelian property). Again, if no further explanation is given, the definition of induced maps and verification of (i)-(ii) is supposed to be obvious. Here the point is that if $\phi : A \longrightarrow B$

is any homomorphism of abelian groups, then $\phi(tA) \subset tB$, yielding the desired induced homomorphism $t\phi : tA \longrightarrow tB$.

Example. Let G be a group. The *abelianization* of G is the quotient group $G_{ab} = G/[G,G]$, where $[G,G]$ is the commutator subgroup, defined as the subgroup generated by all commutators $xyx^{-1}y^{-1}$. By construction, G_{ab} is an abelian group. Then $G \mapsto G_{ab}$ defines a functor from **Grp** to **Ab**. Again, if no further explanation is given, the definition of the induced maps is supposed to be something obvious. In this case there is a little more to check, but it's straightforward: If $\phi : G \longrightarrow H$ is a group homomorphism, then $\phi([G,G]) \subset [H,H]$ and hence we get a well-defined homomorphism $\phi_{ab} : G_{ab} \longrightarrow H_{ab}$ by $\phi_{ab}(g[G,G]) = \phi(g)[H,H]$. This is our induced morphism in **Ab**, and again the verification of (i)-(ii) is trivial.

Example: products. Given two groups G, H we can form the product group $G \times H$. Given homomorphisms $\phi : G_1 \longrightarrow G_2$ and $\psi : H_1 \longrightarrow H_2$, we can form the product homomorphism $\phi \times \psi : G_1 \times H_1 \longrightarrow G_2 \times H_2$ by $(\phi \times \psi)(g,h) = (\phi(g), \psi(h))$. This construction commutes with compositions and takes identities to identities, and so could be said to define a "functor of two variables". But we don't need to introduce any new concept here; as the reader can check, what he have is a ordinary covariant functor from the product category **Grp** $\times$ **Grp** to **Grp**. The same thing works in many familiar categories such as sets, rings, topological spaces, and with the product of two objects replaced by products over an arbitrary index set.

Non-example. We have noted that to check something is a functor is usually a trivial enterprise. On the other hand, one does have to do the checking! For example, if someone says to you "define a functor **Grp** $\longrightarrow$ **Ab** by taking a group G to its center $C(G)$", you should make a note never to trust that person. A group homomorphism need not take centers to centers, so there is no reasonable way to make this assignment a functor.

Here are some simple general types of functors (the definition of induced morphisms, and verification of (i)-(ii), is left to you!).

Example: inclusion functors. If $\mathcal{B}$ is a subcategory of $\mathcal{A}$, then there is an evident inclusion functor $\mathcal{B} \longrightarrow \mathcal{A}$. For example, $i :$ **Ab** $\longrightarrow$ **Grp**.

Example: forgetful functors. In the case of "sets with extra structure", we get functors that simply "forget" some or all of the structure. For example, we have a functor **Grp** $\longrightarrow$ **Set** that just takes a group to its underlying set (and similarly for **Rng**, **Top**, etc.). Similarly **Rng** $\longrightarrow$ **Ab** forgets the ring multiplication but remembers the underlying abelian group. The reader can easily cook up many examples of this type.

4.2 Contravariant functors

The adjective "covariant" introduced at the beginning serves to distinguish functors as above from *contravariant functors*, which we now describe. A contravariant functor $F : \mathcal{A} \longrightarrow \mathcal{B}$ again assigns to each object A of $\mathcal{A}$ an object $F(A)$ of $\mathcal{B}$, but it reverses the direction of morphisms: to each pair of objects A_1, A_2 of $\mathcal{A}$ there is an associated map

$$F : Hom_{\mathcal{A}}(A_1, A_2) \longrightarrow Hom_{\mathcal{B}}(F(A_2), F(A_1)),$$

such that $F(Id_A) = Id_{F(A)}$ and $F(g \circ f) = F(f) \circ F(g)$; note the reversal of order of composition. If the functor F is understood, we sometimes write f^* for $F(f)$. In this notation the composition rule is $(g \circ f)^* = f^* \circ g^*$. Once again, all this is easily understood by looking at a couple of examples.

Example. If V is a vector space over a field F, the *dual* vector space is $V^* := Hom_F(V, F)$. (This "upper star" notation for dual vector spaces is probably where the notation f^* used above originated.) Then $V \mapsto V^*$ defines a contravariant functor $Vect_F \longrightarrow Vect_F$, with the induced maps defined as follows: Given a linear map $f : V \longrightarrow W$, the induced map $f^* : W^* \longrightarrow V^*$ is just precomposition with f, i.e. if $\lambda \in W^*$ then $f^*\lambda = \lambda \circ f$. The formula $(g \circ f)^* = f^* \circ g^*$ is immediate.

Example. If X is a topological space, let $C(X, \mathbb{R})$ denote the ring of continuous real-valued functions on X.[1] Then $X \mapsto C(X, \mathbb{R})$ is a contravariant functor **Top** $\longrightarrow$ **Rng**. As in the preceeding example, the induced maps are given by precomposition: If $f : Y \longrightarrow X$ is a continuous map, and $\lambda : X \longrightarrow \mathbb{R}$ is a continuous function, then $f^*\lambda = \lambda \circ f$. The verification of the axioms for a contravariant functor is, as usual, trivial.

Remark. In fact any contravariant functor can be regarded as a covariant functor, by the following device: If $\mathcal{C}$ is a category, define the *opposite category* $\mathcal{C}^{op}$ to have the same objects as $\mathcal{C}$, but with

$$Mor_{\mathcal{C}^{op}}(X, Y) = Mor_{\mathcal{C}}(Y, X)$$

and the evident composition law. Then a contravariant functor $\mathcal{A} \longrightarrow \mathcal{B}$ is the same thing as a covariant functor $\mathcal{A} \longrightarrow \mathcal{B}^{op}$.

4.3 Functors and isomorphisms; full and faithful functors

The reader may have noticed that so far we haven't stated and proved a single theorem of category theory. Just to demonstrate how simple the beginnings of the subject are, we'll do so now.

Proposition 4.1 *Any functor (covariant or contravariant) takes isomorphisms to isomorphisms.*

Proof: Let $F : \mathcal{A} \longrightarrow \mathcal{B}$ be a covariant functor, and suppose $f : A_1 \longrightarrow A_2$ is an isomorphism in $\mathcal{A}$. Then there exists $g : A_2 \longrightarrow A_1$ such that $g \circ f = Id_{A_1}$ and $f \circ g = Id_{A_2}$. Then

$$F(g) \circ F(f) = F(g \circ f) = F(Id_{A_1}) = Id_{F(A_1)}.$$

Similarly $F(f) \circ F(g) = Id_{F(A_1)}$. Hence $F(f)$ is an isomorphism in $\mathcal{B}$.

[1] In fact it is not merely a ring but an $\mathbb{R}$-algebra, a concept to be discussed later.

Remark: Note that we don't need to prove the contravariant case separately, trivial though this would be. If F were contravariant, we could just replace $\mathcal{B}$ by $(\mathcal{B})^{op}$ to get a covariant functor.

A functor $F : \mathcal{A} \longrightarrow \mathcal{B}$ is *faithful* (resp. *full*) if for all objects A_1, A_2, the map $F : Hom_{\mathcal{A}}(A_1, A_2) \longrightarrow Hom_{\mathcal{B}}(B_1, B_2)$ is injective (resp. surjective). For example, an inclusion functor of a subcategory is faithful by definition, and is full if and only if the subcategory is full. Forgetful functors are always faithful and with trivial exceptions are never full. The abelianization functor is neither faithful nor full (give examples showing this!).

Now, here's another illustration of the simplicity of basic category theory. It is a partial converse to the preceeding proposition.

Proposition 4.2 *Let $F : \mathcal{A} \longrightarrow \mathcal{B}$ be a faithful and full functor, A_1, A_2 objects of A. If $F(A_1)$ is isomorphic to $F(A_2)$, then A_1 is isomorphic to A_2.*

Proof: Suppose $f : F(A_1) \longrightarrow F(A_2)$ is an isomorphism, with inverse g. Since F is full, there are morphisms $h : A_1 \longrightarrow A_2$ and $i : A_2 \longrightarrow A_1$ such that $F(h) = f$ and $F(i) = g$. Then

$$F(i \circ h) = F(i) \circ F(h) = g \circ f = Id_{F(A_1)} = F(Id_{A_1}).$$

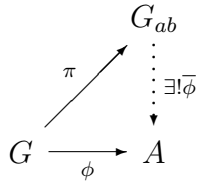
Since F is faithful, $i \circ h = Id_{A_1}$. Similarly $h \circ i = Id_{A_1}$, so h and i are mutually inverse isomorphisms.

5 Universal properties

We don't have a formal definition of a "universal property". It is only an informal term, the significance of which becomes clear after seeing a few examples. One common thread is that if an object has a "universal property", then that property uniquely determines the object, up to a canonical isomorphism. Here the term "canonical" again has no technical meaning; it only implies that the isomorphism in question arises in some natural way from the given data. For a formal approach that covers most of the cases we consider, see the section below on adjoint functors. We will also give a "plain English" interpretation of our universal properties.

For our first example, we consider the abelianization $\pi : G \longrightarrow G_{ab}$ of a group G. We are going to show that this homomorphism has a universal property among all homomorphisms from G to an abelian group.

Proposition 5.1 *If A is an abelian group and $\phi : G \longrightarrow A$ is a homomorphism, then ϕ factors uniquely through G_{ab}:*

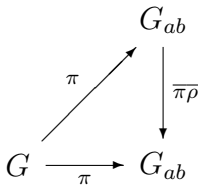


Moreover, this property determines G_{ab} up to a canonical, indeed unique (in a sense to made clear in the proof), isomorphism.

Plain English version: If you want to define a homomorphism from G_{ab} to the abelian group A, it suffices (indeed is equivalent) to define a homomorphism $G \longrightarrow A$.

Proof with discussion. We'll give the proof in considerable detail this time; later on, many such details will be omitted. First of all, the "uniqueness" clause suggests an approach governed by the motto (not to be taken literally!) "uniqueness yields existence": If the darn thing is unique, then its definition should be forced on us, thereby yielding the existence. In the present case the application of our motto is a triviality: Clearly we have no choice but to take $\overline{\phi}(x[G,G]) = \phi(x)$. This works because all commutators are trivial in A, so $[G,G] \subset Ker\,\phi$, and by elementary group theory $\overline{\phi}$ is a well-defined group homomorphism.

It remains to prove the last assertion of the proposition. This part is purely mechanical, once one unravels what the assertion is asserting. Suppose we are given an abelian group H and a homomorphism $\rho : G \longrightarrow H$ satisfying the same property as G_{ab} (i.e., substitute H for G_{ab} and ρ for π in the above commutative triangle). We wish to show that H is isomorphic to G_{ab} in some canonical, "unique" way. Taking $A = H$, $\phi = \rho$ in the diagram above yields a unique homomorphism $\overline{\rho} : G_{ab} \longrightarrow H$. Reversing the roles of H and G_{ab} yields similarly a unique homomorphism $\overline{\pi} : H \longrightarrow G_{ab}$. In fact these two maps are mutually inverse isomorphisms. To see this, note there is a commutative triangle



But this triangle also commutes with $\overline{\pi}\overline{\rho}$ replaced by the identity, hence by uniqueness we must have $\overline{\pi}\overline{\rho} = Id_{G_{ab}}$. Similarly $\overline{\rho}\overline{\pi} = Id_H$, completing the proof.

Another example, perhaps more familiar, concerns products. For this purpose we could take any of the four categories listed at the beginning of this chapter; for variety, let's work in the category of rings. If R_1, R_2 are rings, the product ring is equipped with projection homomorphisms $\pi_i : R_1 \times R_2 \longrightarrow R_i$. The next proposition shows it has a corresponding universal property.

Proposition 5.2 *If R is a ring and $f_i : R \longrightarrow R_i$ is a ring homomorphism for $i = 1, 2$, then there is a unique ring homomorphism $f : R \longrightarrow R_1 \times R_2$ such that $\pi_i f = f_i$. Moreover this property characterizes $R_1 \times R_2$ up to a canonical, indeed unique (in a sense made clear by the proof) isomorphism.*

Plain English version. If you want to define a homomorphism $R \longrightarrow R_1 \times R_2$, it suffices (indeed is equivalent) to define homomorphisms $R \longrightarrow R_1$ and $R \longrightarrow R_2$.

Proof: The first statement is clear. The second follows the formal template laid out in the previous proposition: If S is another ring equipped with homomorphisms $q_i : S \longrightarrow R_i$ having

the same universal property, the universal property of $R_1 \times R_2$ (resp. S) yields a suitably unique homomorphism $R_1 \times R_2 \longrightarrow S$ (resp. $S \longrightarrow R_1 \times R_2$), and applying the universal property two more times (to the two composites) shows that these homomorphisms are mutually inverse isomorphisms.

In the future, we will omit the "unique up to isomorphism" part of the proof entirely. It is also possible to formulate a general principle, based on adjoint functors, that proves all such uniqueness statements at once. This is discussed in the next section, but the discussion there is optional reading.

The universal property of products differs from that of abelianizations in a significant respect: the property of products makes sense in any category whatsoever. In other words, if $\mathcal{A}$ is a category, and A_1, A_2 are objects, we say that the object A is the *(categorical) product* of A_1, A_2 if it is equipped with morphisms $\pi_i : A \longrightarrow A_i$ such that the property above is satisfied. There is no guarantee that A exists, but if it does we denote it $A_1 \times A_2$. In many of the categories we consider, products over any index set obviously exist, by simply taking the set-theoretic product together with the appropriate structure. Products of rings, groups or vector spaces, for instance, have their operations defined componentwise. In the category **Top**, the product topology is designed precisely so that $\prod_{\alpha \in J} X_\alpha$ is the categorical product. Categories in which products fail to exist are easily constructed by simply passing to suitable full subcategories. For example, the category of finite sets has finite products but not infinite products, while the category of sets of cardinality 2 has no products at all.

One more example: An *initial object* (resp. *terminal object*) of a category $\mathcal{C}$ is an object X such that for all objects Y, there is a unique morphism $X \longrightarrow Y$ (resp. $Y \longrightarrow X$). A *null object* is an object that is both initial and terminal. If an object of any of these three types exists, then it is unique up to a unique isomorphism. (Check this!) Simple examples abound: In **Set**, the empty set is initial and the one-point set is terminal; similarly in **Top**. In **Grp**, the trivial group is a (the) null object. In **Rng**, $\mathbb{Z}$ is initial and the zero ring is terminal. In contrast, the full subcategory of **Rng** consisting of the fields has neither initial nor terminal objects (why?).

6 Coproducts and other "dual" constructions

Purely categorical concepts can be formally "dualized" by simply reversing the arrows. For example, consider the definition of an initial object just given. Taking the same definition but with arrows reversed yields the definition of a terminal object. In this sense, the concepts "initial" and "terminal" are dual.

An example of particular importance for us is the concept dual to "product". Recall that if $\mathcal{C}$ is a category, I some index set and X_i a set of objects indexed by I, their product (if it exists) $\prod_{i \in I} X_i$ is an object equipped with morphisms $\pi_j : \prod_{i \in I} X_i \longrightarrow X_j$ and defined by the universal property:

For any object Y and morphisms $f_i : Y \longrightarrow X_i$, there is a unique morphism $f : Y \longrightarrow \prod_{i \in I} X_i$ such that $\pi_i f = f_i$ for all i.

Moreover this property determines the product up to a canonical isomorphism. Now let's just reverse the arrows and define the *coproduct* $\coprod_{i\in I} X_i$ to be an object equipped with morphisms $\alpha_i : X_i \longrightarrow \coprod_{i\in I} X_i$, satisfying the universal property:

For any object Y and morphisms $f_i : X_i \longrightarrow Y$, there is a unique morphism $f : \coprod_{i\in I} X_i \longrightarrow Y$ such that $f \circ \alpha_i = f_i$ for all i.

As usual, the coproduct will be unique if it exists. And as is often the case in category theory, this seemingly abstract, opaque definition becomes concrete and transparent once you look at some simple examples.

Example. In **Set**, coproducts exist and are given by disjoint union. The verification is trivial; in particular there should be no doubt about what the maps α_i are.

Example. In **Top**, coproducts exist and are again given by disjoint union, with the disjoint union topology (i.e. a set is open in $\coprod X_i$ if and only if its intersection with every X_i is open). Again the verification is trivial.

Example. In **Ab**, coproducts exists and are given by *direct sum*, defined as the subgroup of $\prod A_i$ consisting of elements having only finitely many nonzero coordinates. In this case we use the notation $\oplus A_i$ instead of $\coprod A_i$. Here α_i is (once again) the obvious inclusion, and given the f_i's we set $f(\{a_i\}) = \sum f_i(a_i)$, which makes sense since only finitely many of the a_i's are nonzero. Note that a finite coproduct (meaning the index set is finite) is the same as the product.

A similar construction works for vector spaces over a field, and later we will generalize it to modules over a ring.

Life is not always this simple, however. For example, does the category of groups have coproducts? It turns out the answer is yes, but the construction isn't so obvious, even for the coproduct of two groups G, H. Note first of all that the construction given for abelian groups doesn't work here, even if G and H happen to be abelian. In other words, the product $G \times H$ is definitely not the coproduct in **Grp**; try it and see what goes wrong. The coproduct in **Grp** (sticking to just two groups, for simplicity) is classically known as the "free product", denoted $G * H$. Intuitively the construction is fairly simple: you just take formal finite strings of symbols $x_1....x_n$, where each x_i is either in G or H, subject to the equivalence relation that replaces two adjacent symbols lying in the same group by their product, and with multiplication of strings given by concatenation of symbols. It takes some tedious work, however, to make this precise. Free products of groups are important in topology, where they arise in the context of the "fundamental group" of a space. But we will have neither time nor reason to study them here.

Further interesting examples of coproducts will arise later, for example in the category of commutative algebras over a commutative ring. There is also a vast generalization of the product/coproduct type of duality to "limits and colimits", certain special cases of which we will eventually need to consider.

Finally, cheap examples of categories with no coproducts are easily constructed by passing to suitable full subcategories of familiar categories. For example in the category of fields (a full subcategory of **Rng**), clearly there is no coproduct $\mathbb{F}_p \coprod \mathbb{Q}$! (Why?)

7 Natural transformations

Suppose $F, G : \mathcal{A} \longrightarrow \mathcal{B}$ are covariant functors (contravariant functors are treated similarly). A *natural transformation* $T : F \longrightarrow G$ assigns to each object A of $\mathcal{A}$ a morphism $T_A : F(A) \longrightarrow G(A)$ in $\mathcal{B}$, such that for all morphisms $f : A_1 \longrightarrow A_2$ in $\mathcal{A}$, the following diagram commutes:

$$\begin{array}{ccc} F(A_1) & \xrightarrow{F(f)} & F(A_2) \\ \downarrow{\scriptstyle T_{A_1}} & & \downarrow{\scriptstyle T_{A_2}} \\ G(A_1) & \xrightarrow[G(f)]{} & G(A_2) \end{array}$$

A natural transformation is also called a "morphism of functors". The idea is that it provides a morphism from $F(A)$ to $G(A)$ that is so "natural", so completely devoid of arbitrary choices, that it commutes with all morphisms in the above sense. A *natural isomorphism*, or "isomorphism of functors", is a natural transformation T such that for all objects A, T_A is an isomorphism.

Example. Let F be a field, V a vector space over F. Let $V^{**} = (V^*)^*$ denote the double-dual of V. Note the composition of two contravariant functors is covariant, so $V \mapsto V^{**}$ is a covariant functor. Define $T_V : V \longrightarrow V^{**}$ be the linear map defined by $T_V(v)(\lambda) = \lambda(v)$ for $v \in V$, $\lambda \in V^*$. Then T is a natural transformation from the identity functor to V^{**}, i.e. if $A : V \longrightarrow W$ is a linear map, then the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{A} & W \\ \downarrow{\scriptstyle T_V} & & \downarrow{\scriptstyle T_W} \\ V^{**} & \xrightarrow[A^{**}]{} & W^{**} \end{array}$$

The proof is an immediate check, once you unravel the definition of A^{**}: If $\mu : V^* \longrightarrow F$ is an element of $(V^*)^*$, and $\lambda \in W^*$, then $(A^{**}(\mu))(\lambda) = \mu(\lambda \circ A)$.

It is clear that T_V is always injective, and hence is an isomorphism when V is finite-dimensional. Thus for finite-dimensional V, T is a natural isomorphism. (But T_V is never surjective if V is infinite-dimensional, a recommended exercise.)

This example should be contrasted with what happens for the dual itself. If V is finite-dimensional, then $V \cong V^*$ as vector spaces. But there is no natural isomorphism; to get an isomorphism one must arbitrarily choose a basis $\{e_i\}$ for V, take e_i^* to be the dual basis,

and use the unique linear transformation taking e_i to e_i^* (or some equivalent procedure). Remember that for $v \in V$ there is no "dual element" v^*. There are only dual *bases*.

Example. For an abelian group A and $n \geq 0$, we set $A[n] = \{a \in A : na = 0\}$. Consider the functors $\mathbf{Ab} \longrightarrow \mathbf{Ab}$ given by $A \mapsto Hom(\mathbb{Z}/n, A)$ and $A \mapsto A[n]$. I claim these two functors are isomorphic. In fact it is easy to check that the map $T_A : Hom(\mathbb{Z}/n, A) \longrightarrow A[n]$ given by $f \mapsto f(1)$ is an isomorphism, and completely trivial to check that the naturality diagram commutes. In fact even before doing the latter check, we are very confident that it will work, because of the "natural" way in which the transformation is defined.

8 Representable functors

For any fixed object X of a category $\mathcal{C}$ there is a covariant (resp. contravariant) functor $\mathcal{C} \longrightarrow \mathbf{Set}$ given by $A \mapsto Hom(X, A)$ (resp. $A \mapsto Hom(A, X)$). The induced morphisms are given by postcomposition in the covariant case and precomposition in the contravariant case. If the category is "enriched" as described earlier, the Hom sets may have some extra structure and we get a functor to our more structured category rather than just to **Set**. As usual, the way to understand this onslaught of verbiage is to look at an example: Take $\mathcal{C} =$**Ab**; then for abelian groups X, A we know that $Hom(X, A))$ is itself an abelian group, and $A \mapsto Hom(X, A)$ is a functor to abelian groups (and similarly for the contravariant version). We call X the *representing object* of the functor.

We next give two useful properties of such functors. We'll take the contravariant case to illustrate. Fix objects Y, Z and consider the functors $Hom(-, Y)$, $Hom(-, Z)$. Any morphism $f : Y \longrightarrow Z$ yields a natural transformation $Tf : Hom(-, Y) \longrightarrow Hom(-, Z)$ by postcomposition with f.

Lemma 8.1 *Tf is a (natural) isomorphism if and only if f is an isomorphism.*

Proof: The "if" is clear. Now suppose $Tf : Hom(X, Y) \longrightarrow Hom(X, Z)$ is a bijection for all X. Taking $X = Z$, it follows from the surjectivity that there is a $g : Z \longrightarrow Y$ such that $f \circ g = Id_Z$. Then $f(gf) = (fg)f = f$. But also $f(Id_Y) = f$, so by the injectivity $gf = Id_Y$. Hence f, g are mutually inverse isomorphisms.

In fact every natural transformation of such functors comes from a unique morphism of the representing objects.

Lemma 8.2 *Suppose $T : Hom(-, Y) \longrightarrow Hom(-, Z)$ is a natural transformation. Then there is a unique morphism $f : Y \longrightarrow Z$ such that $T = Tf$.*

Proof: If $T = Tf$, then $T(Id_Y) = f$. Thus f is unique if it exists. For the existence, we are then forced to *define* $f = T(Id_Y)$. To see that this works, we must show that for any morphism $g : X \longrightarrow Y$, we have $T(g) = f \circ g$. Since T is a natural transformation, we have a commutative diagram

$$\begin{array}{ccc} Hom\,(Y,Y) & \xrightarrow{T} & Hom\,(Y,Z) \\ \Big\downarrow{\scriptstyle g^*} & & \Big\downarrow{\scriptstyle g^*} \\ Hom\,(X,Y) & \xrightarrow[T]{} & Hom\,(X,Z) \end{array}$$

Now start with Id_Y in the upper left and follow it both ways around the diagram. Going around the left and bottom yields $T(g)$. Going around the top and right yields $f \circ g$. This completes the proof, vindicating once again our motto "uniqueness yields existence".

A functor $F : \mathcal{C} \longrightarrow \mathbf{Set}$ is *representable* if it isomorphic as a functor to $Hom\,(X, -)$ in the covariant case (similarly for the contravariant case). We say that F is *represented by* X. This is a very special property to have, and indeed there are some big theorems (in topology, for example) asserting that certain functors are representable. What's nice about representable functors is that often they can be understood by understanding one object, namely the representing object X. In this course we probably won't have any deep examples, but even in simple cases it can help to know that your functor is representable.

Example. Consider the covariant functor $\mathbf{Ab} \longrightarrow \mathbf{Ab}$ given by $A \mapsto A[n] := \{a \in A : na = 0\}$. In the previous section we showed it was isomorphic to $Hom\,(\mathbb{Z}/n, A)$; now we can express this by saying that $A[n]$ is representable, with representing object $\mathbb{Z}/n$.

Example: Consider the covariant functor $\mathbf{G\text{-}set} \longrightarrow \mathbf{Set}$ given by $S \mapsto S^G$. This functor is represented by the trivial one-point G-set, again a trivial check. More generally if $H \subset G$ is a fixed subgroup, the functor $\mathbf{G\text{-}set} \longrightarrow \mathbf{Set}$ given by $S \mapsto S^H$ is represented by G/H.

9 Adjoint functors

Adjoint functors are everywhere. You can hardly walk down the street without running into one. They often travel in disguise, however, and you may have to look closely to recognize them. In particular, almost all of the "universal properties" we consider can be formulated in adjoint functor terms.

Definition. Let $\mathcal{A}$, $\mathcal{B}$ be categories, and let $F : \mathcal{A} \longrightarrow \mathcal{B}$, $G : \mathcal{B} \longrightarrow \mathcal{A}$ be covariant functors. Then F is *left adjoint* to G, or equivalently G is *right adjoint* to F, if there is a natural bijection

$$Hom_{\mathcal{B}}(F(A), B) \xrightarrow{\cong} Hom_{\mathcal{A}}(A, G(B))$$

where A, B are objects of $\mathcal{A}$, $\mathcal{B}$ respectively. Here "natural" means natural in both variables, i.e. if one variable is held fixed we have a natural transformation of set-valued functors in the other variable. Let's make this explicit. To simplify the writing, from here on we use the following notation: If X, Y are objects of a category $\mathcal{C}$, we set $[X, Y] = Hom_{\mathcal{C}}(X, Y)$. There will be no ambiguity, because the category $\mathcal{C}$ in question will be implicitly determined by the objects X, Y.

Naturality in B means that for every morphism $g : B_1 \longrightarrow B_2$ in $\mathcal{B}$, the following diagram commutes:

$$\begin{array}{ccc} [F(A), B_1] & \xrightarrow{\cong} & [A, G(B_1)] \\ \downarrow{\scriptstyle g_*} & & \downarrow{\scriptstyle G(g)_*} \\ [F(A), B_2] & \xrightarrow[\cong]{} & [A, G(B_2)] \end{array}$$

Here the horizontal arrows are the adjunction bijections given by hypothesis, while the vertical arrows are just postcompositions. Similarly, naturality in A means that the following diagram commutes:

$$\begin{array}{ccc} [F(A_2), B] & \xrightarrow{\cong} & [A_2, G(B)] \\ \downarrow{\scriptstyle F(f)^*} & & \downarrow{\scriptstyle f^*} \\ [F(A_1), B] & \xrightarrow[\cong]{} & [A_1, G(B)] \end{array}$$

Here the vertical arrows are precompositions.

Now, let's look at a few of the hundreds of examples.

Example: abelianization. The abelianization functor $\mathbf{Grp} \longrightarrow \mathbf{Ab}$ is left adjoint to the inclusion functor $i : \mathbf{Ab} \longrightarrow \mathbf{Grp}$, and indeed this assertion is equivalent to the universal property established earlier. The universal property says that given any homomorphism $G \longrightarrow A$ (G a group, A an abelian group), determines a unique homomorphism $G_{ab} \longrightarrow A$; in other words, it establishes a bijection (in fact an isomorphism of abelian groups, although for the moment I only care about the bijection)

$$Hom_{Ab}(G_{ab}, A) \xrightarrow{\cong} Hom_{Grp}(G, i(A))$$

given by sending a homomorphism $G_{ab} \longrightarrow A$ to the composite $G \longrightarrow G_{ab} \longrightarrow A$.
To conclude that these are adjoint functors, what remains to be shown is that the above bijection is natural. As is typical, this is a trivial check. Consider, for example, naturality in A. This says that if $\phi : A \longrightarrow B$ is a homomorphism of abelian groups, then the following diagram commutes:

$$\begin{array}{ccc} Hom\,(G_{ab}, A) & \xrightarrow{\phi_*} & Hom\,(G_{ab}, B) \\ \downarrow & & \downarrow \\ Hom\,(G, A) & \xrightarrow[\phi_*]{} & Hom\,(G, B) \end{array}$$

where the vertical arrows are precomposition along $G \longrightarrow G_{ab}$. It is then immediate that the diagram commutes. The naturality in G is equally easy. From here on all checks of this kind will be left to the reader.

Example: products Let $\mathcal{C}$ be any category with products, such as **Set**, **Grp**, etc. Then the product functor $P : \mathcal{C} \times \mathcal{C} \longrightarrow \mathcal{C}$ is right adjoint to the diagonal functor $\Delta : \mathcal{C} \longrightarrow \mathcal{C} \times \mathcal{C}$ given by $\Delta(X) = (X, X)$. Here again this assertion is equivalent to the universal property given earlier, since the evident adjunction isomorphism

$$Hom_{\mathcal{C}\times\mathcal{C}}(\Delta(X), (Y, Z)) \cong Hom_{\mathcal{C}}(X, Y \times Z)$$

is doing nothing more than identify a pair of morphisms $X \longrightarrow Y$, $X \longrightarrow Z$ (on the left above) with a morphism $X \longrightarrow Y \times Z$ (on the right). The analogous adjoint functor relationship holds for products indexed by an arbitrary set.

Example: initial/terminal objects. Since initial and terminal objects are so easy to understand directly, it may seem like overkill to identify them in terms of adjoint functors. But we'll do it anyway, just to underscore the ubiquity of the concept. We take initial objects to illustrate. Let **Triv** denote the *trivial category*, i.e. the category with only one object x and with Id_x the only morphism $x \longrightarrow x$. Then for any category $\mathcal{C}$ there is a unique functor $t : \mathcal{C} \longrightarrow$ **Triv**. To say that $\mathcal{C}$ has an initial object is the same thing as saying that this functor has a left adjoint $i : Triv \longrightarrow \mathcal{C}$. To see this, suppose we have such an adjoint, and consider the adjunction isomorphism

$$Hom\,(i(x), C) \cong Hom\,(x, t(C)).$$

Since the righthand side has a single element, we conclude that the lefthand side also consists of a single element, which says precisely that $i(x)$ is an initial object. The reverse implication is equally trivial; it is not very difficult to prove naturality when the Hom-sets in question have a single element!

Many of the most important examples will appear gradually as the course proceeds. For general reference we list a few here, even though most of them will only make sense later on.

- If G is a group and $H \subset G$ is a subgroup, the induction functor **H-set** $\longrightarrow$ **G-set** (given by $X \mapsto G \times_H X$) is left adjoint to the restriction functor.
- The free R-module functor **Set** $\longrightarrow$ **R-mod** is left adjoint to the forgetful functor.
- If R is a commutative ring, the group algebra functor **Grp** $\longrightarrow$ **R-alg** given by $G \mapsto RG$ is left adjoint to the group of units functor $R \mapsto R^{\times}$.

We next want to show that adjoint functors are unique up to isomorphism of functors. To this end it is convenient to introduce a "category of squares", the definition of which is exactly analogous to that of the category of morphisms, but with a square of morphisms replacing a single morphism (the latter can be thought of as an edge). Given any category $\mathcal{C}$, the category of square diagrams $\mathcal{S} = \mathcal{S}_{\mathcal{C}}$ has objects the diagrams

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D \end{array}$$

whose vertices are objects of $\mathcal{C}$ and whose (edge) arrows are morphisms. The diagrams are *not* assumed to commute. A morphism of such diagrams is the obvious thing, namely a cubical diagram that I am too lazy to draw and so will only describe: Our cube has four vertical faces and two horizontal faces. Put the source diagram on the back vertical face and the target diagram at the front. The four horizontal edges of the cube are morphisms $A_1 \longrightarrow A_2$, $B_1 \longrightarrow B_2$ etc., the source being labeled with subscript 1 and the target with subscript 2. We require that the four new squares, that are formed by the top, bottom, left and right faces of the cube, commute. For example, the top face of the cube is a commutative square

$$\begin{array}{ccc} A_1 & \longrightarrow & B_1 \\ \downarrow & & \downarrow \\ A_2 & \longrightarrow & B_2 \end{array}$$

It is clear that this makes $\mathcal{S}$ into a category, and like any category it has isomorphisms. We then make the simple

Observation. If a square (i.e. an object) in $\mathcal{S}$ is commutative, then so is any square isomorphic to it.

Proposition 9.1 *Let $F : \mathcal{A} \longrightarrow \mathcal{B}$ be a covariant functor, and suppose F has a right adjoint $G : \mathcal{A} \longrightarrow \mathcal{B}$. Then G is unique up to isomorphism of functors. Left adjoints are similarly unique up to isomorphism.*

Proof: Suppose G_1, G_2 are both right adjoints of F. For any fixed object B of $\mathcal{B}$, the contravariant functors $[-, G_1B]$, $[-, G_2B]$ represent the same functor (up to isomorphism) on $\mathcal{A}$, namely $[F(-), B]$. Hence by Lemma 8.2, there is an isomorphism $g = g_B : G_1B \longrightarrow G_2B$ characterized by the property that for all objects A of $\mathcal{A}$, the following **Diagram 1** commutes:

$$\begin{array}{ccc} [F(A), B] & \xrightarrow{=} & [F(A), B] \\ \downarrow{\scriptstyle \phi_1} & & \downarrow{\scriptstyle \phi_2} \\ [A, G_1B] & \xrightarrow[g_*]{} & [A, G_2B] \end{array}$$

Here ϕ_1, ϕ_2 are the adjunction isomorphisms that are given by hypothesis. We could collapse the top edge of the diagram to make a commutative triangle, but for later reference we leave it as is.

What remains to be shown is that the isomorphism g is a natural transformation. In other words, we must show that for all morphisms $h : B \longrightarrow C$ in $\mathcal{B}$, the following diagram commutes:

$$\begin{array}{ccc} G_1B & \xrightarrow{g} & G_2B \\ \downarrow{\scriptstyle h_*} & & \downarrow{\scriptstyle h_*} \\ G_1C & \xrightarrow[g]{} & G_2C \end{array}$$

Now in any category $\mathcal{C}$, two morphisms $Y \longrightarrow Z$ agree if and only if the induced morphisms $[X, Y] \longrightarrow [X, Z]$ agree for all objects X. Hence a diagram in $\mathcal{C}$ commutes if and only if the diagram of sets obtained by applying the functor $[X, -]$ commutes for all X. Applying this principle to the preceeding diagram, what we need to show is that for all objects A of $\mathcal{A}$, the induced diagram of sets

$$\begin{array}{ccc} [A, G_1B] & \longrightarrow & [A, G_2B] \\ \downarrow & & \downarrow \\ [A, G_1C] & \longrightarrow & [A, G_2C] \end{array}$$

commutes. By the *observation* above, it suffices to show that it is isomorphic to the square

$$\begin{array}{ccc} [F(A), B] & \xrightarrow{=} & [F(A), B] \\ \downarrow{\scriptstyle h_*} & & \downarrow{\scriptstyle h_*} \\ [F(A), C] & \xrightarrow[=]{} & [F(A), C] \end{array}$$

which obviously commutes. To this extend we construct the cube whose top face is the commutative **Diagram 1** and whose bottom face is the analogous commutative square with B replaced by C. This defines the desired isomorphism, provided that the left and right faces are also commutative squares. But the left face is precisely the square expressing naturality in B of the adjoint isomorphism $[A, G_1B] \cong [F(A), B]$, and hence is commutative, while the right face is the corresponding square for G_2. This completes the proof.

10 Coming attractions

To be continued. Equivalence of categories, limits and colimits, and other fun stuff.